

INTELLIGENT ROUTE PLANNER WITH MULTI-MODAL TRANSPORT

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Abstract— Urban journeys increasingly require users to combine walking, cycling, bus, and metro services. Conventional route planners often emphasize shortest distance or minimum travel time and may not adequately represent waiting time, transfers, schedules, cost, comfort, and user preferences. This paper presents an *Intelligent Route Planner with Multi-Modal Transport* that models a journey as a multi-modal transportation graph and evaluates alternatives using Dijkstra's Algorithm, A* search, and Anytime Weighted A*. The framework incorporates time-dependent costs arising from departure time, traffic conditions, public-transport schedules, waiting time, and transfer time. A multi-objective ranking layer considers travel time, transportation cost, walking distance, number of transfers, and comfort, allowing routes to be classified as fastest, cheapest, comfortable, or balanced. An interactive map layer is proposed to visualize the complete journey. The paper describes the architecture and algorithmic framework of the project; numerical performance claims are not introduced where implementation measurements are unavailable.

Keywords— Multimodal transportation, route planning, Dijkstra, A*, Anytime Weighted A*, time-dependent routing, multi-objective optimization.

I. INTRODUCTION

Urban mobility frequently involves more than one transportation mode. A passenger may walk to a bus stop, travel by bus, transfer to a metro, and walk again at the destination. Selecting a suitable combination is more complex than finding a single shortest path because route quality depends on departure time, public-transport schedules, waiting, traffic, transfers, cost, and personal preferences.

Dijkstra's algorithm provides a fundamental minimum-cost path method, while A* improves search by using a heuristic estimate of the remaining cost. However, public transportation introduces time-dependent effects: a scheduled vehicle may have departed, a transfer may require additional waiting, and road congestion may change travel time. Therefore, a practical multimodal planner needs to combine graph search with temporal and user-specific information.

The proposed system integrates walking, cycling, bus, and metro in a common graph and combines Dijkstra, A*, and Anytime Weighted A* with time-dependent evaluation and multi-objective ranking. Instead of returning only one shortest route, the planner compares alternatives according to speed, cost, comfort, walking distance, and transfers.

The main contributions are: (1) a unified multimodal route representation; (2) integration of classical and heuristic shortest-path algorithms; (3) time-dependent evaluation of traffic, schedules, waiting, and transfers; (4) preference-based multi-objective ranking; and (5) interactive visualization of the complete journey.

II. RELATED WORK

Dijkstra introduced a shortest-path method for weighted graphs that remains a fundamental basis for network routing [1]. Hart, Nilsson, and Raphael established the theoretical foundation of A* heuristic search, where a cost-so-far term is combined with an estimate of the remaining cost [2].

Time-dependent routing is important because transportation costs vary with departure time. Verbas et al. proposed a Time-Dependent Intermodal A* algorithm for networks containing transit, walking, and vehicular links, including transfer penalties and bounds on waiting and walking [3].

Aifadopoulou, Ziliaskopoulos, and Chrisohoou developed a multiobjective optimum-path approach for passenger pretrip planning in multimodal networks, considering delays, terminal switching, mode compatibility, and user preferences [4]. Potthoff and Sauer studied fast multimodal journey planning for multiple criteria, demonstrating the importance of optimizing beyond a single arrival-time criterion [5].

These works motivate the proposed combination of multimodal graph modeling, time-dependent routing, heuristic search, and preference-based route ranking. The present project organizes these concepts into a practical architecture suitable for an urban route-planning prototype.

III. PROBLEM STATEMENT AND OBJECTIVES

The problem is to identify an efficient route between a source and destination when the journey may use multiple transportation modes and route quality depends on time, cost, waiting, transfers, and user preferences. A geographically short path may not be optimal when a scheduled service is unavailable or when a user prioritizes lower cost or reduced walking.

Objectives: (1) represent walking, cycling, bus, and metro in a unified graph; (2) apply Dijkstra for minimum-cost routing; (3) apply A* and Anytime Weighted A* for efficient search; (4) model traffic, schedules, waiting, and transfer time; (5) rank routes using time, cost, walking distance, transfers, and comfort; and (6) visualize route alternatives on an interactive map.

IV. PROPOSED SYSTEM ARCHITECTURE

The system uses a layered architecture. User input contains source, destination, departure time, and preferences. The presentation layer provides search and map interfaces, while the application/API layer manages routing requests. Transportation data are converted into a multimodal graph. Search algorithms generate candidate routes, the time-dependent engine updates their costs, and the ranking layer selects alternatives for visualization.

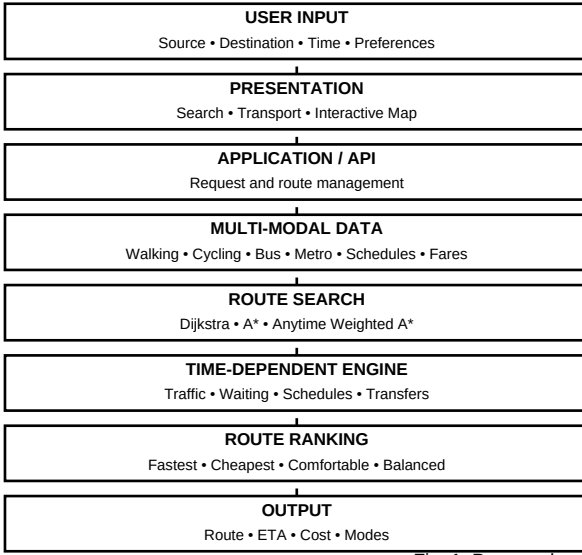


Fig. 1. Proposed architecture of the Intelligent Route Planner.

Nodes represent locations, stops, stations, or transfer states, while edges represent movement using a particular mode. Transfer edges connect compatible modes. This representation permits a journey to move from walking to bus, bus to metro, cycling to transit, or other supported combinations.

V. ROUTE MODEL AND COST FUNCTION

Let the transportation network be represented by a directed graph $G=(V,E)$, where V is the set of locations or transport states and E is the set of feasible movements. Each edge has attributes such as mode, distance, travel time, monetary cost, and transfer information. Scheduled services additionally have availability conditions.

A conceptual time-dependent edge cost is represented as:

$$C(e,t) = w_t T(e,t) + w_c C_{\text{money}}(e) + w_w W(e,t) + w_x X(e) + w_m M(e)$$

where T is travel time, C_{money} is monetary cost, W is waiting time, X is transfer-related cost, M is a comfort penalty or score, and the w terms are user-selected weights. The weights can be adjusted for speed, cost, comfort, or a balanced preference.

For a complete route R , the system aggregates the criteria across its edges and ranks feasible routes. Thus, the same spatial path can receive different evaluations for different departure times or user preferences.

VI. SEARCH ALGORITHMS

A. Dijkstra's Algorithm. Dijkstra repeatedly selects the unvisited node with the smallest known cost and relaxes outgoing edges. It is suitable for non-negative edge costs and provides a baseline minimum-cost solution.

B. A* Algorithm. A* evaluates a node using $f(n)=g(n)+h(n)$, where $g(n)$ is the accumulated cost and $h(n)$ is a heuristic estimate of the remaining cost. A suitable geographical heuristic can guide the search toward the destination while avoiding unnecessary exploration.

$$f(n) = g(n) + h(n)$$

C. Anytime Weighted A*. Anytime Weighted A* uses a weighted heuristic to obtain a good solution quickly and can continue searching to improve solution quality. This is useful when rapid initial route generation is important but additional computation can improve the result.

VII. TIME-DEPENDENT MULTI-MODAL ROUTING

A public-transport edge does not always have a constant effective cost. A passenger arriving after a scheduled bus departure must wait for the next service. Road travel time may also increase during peak traffic. Transfers introduce walking and interchange time. The proposed time-dependent engine evaluates each edge using the user's arrival time at that point in the journey.

For scheduled services, waiting time can be represented by the difference between the next available departure and the user's arrival at the stop or station. The resulting arrival time is then propagated to the next edge. This allows the planner to distinguish between journeys that use the same spatial path but begin at different times.

VIII. MULTI-OBJECTIVE ROUTE RANKING

Route	Objective	Preference
Fastest	Minimize total travel time	Time-sensitive
Cheapest	Minimize monetary cost	Budget-conscious
Comfortable	Reduce walking, transfers and discomfort	Convenience-oriented
Balanced	Trade off time, cost and comfort	General-purpose

Table I. Example route-ranking categories.

A user emphasizing speed can assign a larger weight to travel time, while a cost-sensitive user can emphasize fare. Comfort-oriented routing can penalize excessive transfers or walking. The design can use weighted aggregation for a simple implementation and can later be extended to Pareto-based multi-objective optimization.

IX. SYSTEM WORKFLOW

The workflow begins with source, destination, departure time, and preferences. Relevant transport data are assembled, a multimodal graph is constructed, and the selected search algorithm generates feasible paths. The time-dependent engine evaluates waiting and travel

conditions. Candidate routes are then ranked and displayed with their modes, estimated time, cost, and transfer information.

Step	Stage	Operation
1	User Input	Source, destination, departure time, preferences
2	Data Preparation	Transport network, schedules, fares and map information
3	Graph Construction	Multimodal nodes, edges and transfer links
4	Route Search	Dijkstra / A* / Anytime Weighted A*
5	Time Evaluation	Traffic, schedules, waiting and transfer time
6	Route Ranking	Fastest, cheapest, comfortable and balanced
7	Visualization	Interactive map, ETA, cost and modes

Table II. Proposed route-planning workflow.

X. DISCUSSION AND EVALUATION PLAN

The project source document defines the proposed algorithms and architecture but does not provide completed experimental measurements or a benchmark dataset. Therefore, this paper does not claim numerical accuracy, speed-up, or optimality values that have not been measured. A complete implementation should be evaluated using representative source-destination queries under different departure times and preference settings.

Useful evaluation metrics include computation time, number of explored nodes, route cost, total travel time, walking distance, number of transfers, and quality of the returned route. Dijkstra, A*, and Anytime Weighted A* can be compared on identical graph instances. Time-dependent routing can be tested during peak and non-peak conditions, while the ranking layer can be tested with speed-, cost-, comfort-, and balanced preferences.

XI. CONCLUSION

This paper presented an intelligent route-planning framework for multi-modal urban journeys. The proposed system combines walking, cycling, bus, and metro services within a common graph and uses Dijkstra's Algorithm, A* search, and Anytime Weighted A* to generate suitable routes. Time-dependent evaluation adds traffic, schedules, waiting, and transfer considerations that are absent from a purely static shortest-path model.

The multi-objective ranking layer allows route alternatives to be classified as fastest, cheapest, comfortable, or balanced according to user preferences. The architecture separates transportation data, route search, temporal evaluation, ranking, and visualization, making the system suitable for future expansion. The framework therefore provides a practical foundation for intelligent urban route planning.

XII. FUTURE WORK

Future work can integrate live traffic feeds and public-transport APIs, dynamic rerouting, weather-aware routing, accessibility constraints, carbon-emission estimation, ride-sharing and additional transport modes. Personalized learning can also be used to adapt route recommendations to recurring user preferences. These extensions can transform the prototype into a real-time intelligent urban mobility platform.

XIII. REFERENCES

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